

- 6 A fast moving neutron with an initial velocity u has a head-on elastic collision with a stationary proton. After the collision, the velocity of the neutron is v and that of the proton is w .

Taking the masses of the neutron and proton to be equal, which of the following statements is **incorrect**?

- A Since collision is elastic, it shows that $u + v = w$.
- B The proton and the neutron move off in opposite directions with equal speeds.
- C By considering kinetic energies of the particles, it can be shown that $u^2 = v^2 + w^2$.
- D The speed of the proton after the collision is the same as that of the neutron before the collision.

- 7 A car is travelling on a hump with a radius of curvature of 30 m as shown in figure below. The car